

Matthew Weinert

SOFTWARE ENGINEER

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RESOURCES

<https://www.mattweinert.com> <https://github.com/MattRayWeinert> <https://www.youtube.com/watch?v=HvO5HcZntn8>

PROFESSIONAL EXPERIENCE

BNY

February 2023 - Current

Senior Associate - Full Stack App Developer

Technical Experience

- Languages: Java, TypeScript, JavaScript, SQL
- Frameworks: Spring Boot, Angular, Ionic
- AI & Automation: Enterprise AI Platforms, Agentic AI, AI Workflow Automation, LLM Integration, Prompt Engineering, AI Agents, MCP
- DevOps & Infrastructure: GitLab CI/CD, Docker, Kubernetes, AppEngine, Git
- Databases: Oracle
- Testing: Karma
- Tools: Jira, Confluence

Responsibilities

- Designed and delivered end-to-end AI-powered applications using Eliza, BNY's enterprise AI platform, enabling data-driven financial insights and intelligent workflow automation across multiple business organizations.
- Architected and implemented production-grade AI agent solutions that automated complex business workflows, improving user productivity by 100%+ while significantly reducing manual effort.
- Built executive-facing AI applications that aggregate, enrich, and analyze financial and operational data, enabling real-time strategic decision-making for senior leadership.
- Engineered full-stack applications using Angular, TypeScript, Spring Boot, Java 17, and MySQL, integrating AI inference, backend services, and intuitive user interfaces.
- Developed scalable REST APIs and backend services using Spring Boot and Java 17 supporting enterprise AI applications.
- Designed and maintained CI/CD pipelines using GitLab, Docker, Kubernetes, and BNY's AppEngine deployment platform, integrating enterprise secret management into automated deployment workflows while resolving complex DevOps, deployment, and production issues across containerized microservices to improve system stability, deployment reliability, and release efficiency.
- Collaborated with product managers, data engineers, architects, and executive stakeholders to translate complex business problems into scalable, production-ready AI solutions.
- Presented AI initiatives, architectural designs, and platform capabilities to directors and executive leadership, influencing technology roadmaps and accelerating enterprise AI adoption.
- Co-invented and developed a patented software solution enabling scenario-based financial modeling and predictive analytics to enhance enterprise strategic planning.
- Mentored and led two software engineering intern teams through architecture, implementation, and Agile development; one team earned first place in BNY's internship innovation competition.
- Interviewed 100+ software engineering candidates, contributing to hiring decisions and maintaining engineering quality standards across multiple teams.
- Contributed to cross-platform mobile application development using the Ionic Framework and Angular, extending enterprise applications to mobile users.

4c Strategies

May 2022 - February 2023

Software Engineer

Technical Experience

- Languages: Java, TypeScript, SQL
- Frameworks: Spring Boot, Angular
- Databases: MySQL
- Testing: Karma, JUnit
- DevOps & Tools: Git, GitLab, Jira, Confluence

Responsibilities

- Designed, architected, and developed a full-stack application from the ground up supporting the U.S. Army's portable Synthetic Training Environment (STE), delivering scalable capabilities for virtual training and simulation.
- Engineered modern web applications using Angular, TypeScript, Spring Boot, Java, and MySQL, building responsive user interfaces and scalable backend services for mission-critical defense systems.
- Maintained, modernized, and enhanced the Wargaming (WG) platform used by the U.S. Marines, implementing new features, resolving defects, and improving application performance and usability.
- Collaborated with cross-functional engineering teams, product owners, and government stakeholders to translate complex operational requirements into production-ready software solutions.
- Delivered high-quality software through Agile Scrum methodologies, consistently meeting bi-weekly sprint commitments and release milestones.
- Led the creation and maintenance of software architecture, system design, and technical documentation, improving knowledge sharing and accelerating onboarding across development teams.
- Conducted peer code reviews and developed comprehensive unit tests, improving code quality, maintainability, and long-term reliability.
- Participated in system design discussions, technical planning, and feature estimation, contributing to scalable application architecture and engineering best practices.
- Investigated and resolved complex software defects while providing ongoing support for deployed applications throughout the software development lifecycle.
- Collaborated closely with QA engineers to identify, reproduce, and resolve defects, ensuring stable production releases and a high-quality user experience.

Phoenix Defense Jr. Software Engineer

May 2021 - May 2022

Technical Experience

- Languages: Java, TypeScript, SQL
- Frameworks: Spring Boot, Angular
- Platforms: Android Team Awareness Kit (ATAK), Android Studio, Google Maps SDK
- Databases: MySQL
- Tools: Git, Bitbucket, Jira, Confluence

Responsibilities

- Led engineering efforts for an IoT mesh networking platform by developing a Land Navigation application integrated with the Android Team Awareness Kit (ATAK), enabling decentralized communication and resilient situational awareness during tactical field operations.
- Designed and implemented a distributed mesh networking service that communicated over local peer-to-peer networks, propagating location data across intermediary nodes to provide reliable user tracking without requiring centralized network infrastructure.
- Developed geospatial capabilities by integrating Google Maps with ATAK, enabling real-time visualization of personnel locations, route awareness, and live operational data within a shared tactical map interface.
- Architected and developed a full-stack enterprise asset management application used to track hardware inventory across dozens of military installations and training sites, providing centralized visibility into fielded equipment and lifecycle management.
- Designed and implemented a role-based access control (RBAC) system with entitlement-based permissions, enabling secure management of hardware assets, inventory updates, and administrative operations across multiple user roles.
- Built automated warranty lifecycle management features, including scheduled background services that monitored expiring warranties and generated proactive email notifications, reducing manual oversight and improving equipment readiness.
- Developed backend REST APIs and business services using Spring Boot and Java, supporting inventory management, asset lifecycle tracking, notification workflows, and secure data operations.
- Engineered responsive web applications using Angular and TypeScript, delivering intuitive interfaces for inventory management, reporting, and administrative workflows.
- Collaborated with cross-functional engineering teams and government stakeholders in an Agile environment to translate operational requirements into scalable software solutions while consistently delivering sprint commitments.
- Mentored and onboarded software engineers through technical guidance, code reviews, and knowledge sharing, accelerating team productivity and development best practices.

Dignitas Technologies Software Engineer Intern

July 2019 - April 2020

Technical Experience

- Languages: Java, SQL, HTML, CSS
- Frameworks: Google Web Toolkit (GWT)
- Version Control: Git, TortoiseSVN
- Tools: Jira, Confluence

Responsibilities

- Developed a web-based application that ingested, transformed, and integrated data from multiple disparate sources to support military training simulations, enabling more realistic and data-driven operational scenarios.
- Designed and implemented backend services and application features using Java, SQL, and Google Web Toolkit (GWT), supporting scalable data processing and interactive web-based simulation tools.
- Built data ingestion and validation workflows to normalize incoming datasets, improving data consistency and reliability for downstream simulation systems.
- Collaborated with software engineers, QA analysts, and government stakeholders in an Agile Scrum environment to deliver production-ready features through iterative sprint cycles.
- Participated in application design, feature implementation, and defect resolution while contributing to sprint planning, technical discussions, and release activities.
- Performed field-based system validation and quality assurance testing in operational training environments, identifying defects, documenting issues in Jira, and verifying software fixes prior to production deployment.
- Investigated and resolved software defects through root cause analysis, improving application stability, usability, and overall system reliability.
- Contributed to source control management, code reviews, and software configuration management using Git and TortoiseSVN, following established development best practices.

TECHNICAL EXPERIENCE

- Version controls: *Git Bash, Github, GIT, TortiseSVN, BitBucket*
- Editors: *Eclipse, IntelliJ, Webstorm, Visual Studio, VS Code, NetBeans*
- Operating Systems: *Windows, Linux, Unix*
- Databases/Languages: *SQL, SQLite, MongoDB*
- Backend: *Tomcat, SpringBoot, .NET, Node, Express, GraphQL, REST*
- Frontend: *React, Javascript, Typescript, Angular, Materials, Bootstrap, RXJS*
- A.I. tools & related libraries: *OpenCV, Numpy, Pandas*
- Workflows: *Jira, Confluence, Gitlab*
- Other: *AWS Amplify, AWS BeanStalk, AWS RDS, HostGator, Heroku*

EDUCATION

University of Central Florida (UCF)

B.S. Computer Engineering

- Area of focus in college: Computer Engineering with a special interest in Software development and Computer Sciences, which enabled a career and passion for learning technologies

ASSETS

- *Active Security Clearance*
- *Bachelors in Computer Engineering*
- *Associates Degree*

PROGRAM LANGUAGES

C, C#, Java, TypeScript, JavaScript, Kotlin, Python, PHP, CSS, HTML

PERSONAL PROJECTS

A.I. "Smart-Hub" Security System

Technical Experience

- Languages: Python, MicroPython
- Frameworks/Libraries: PyQt5, OpenCV
- Hardware: Raspberry Pi, Embedded Systems, Sensors
- Design Tools: AutoDesk Eagle CAD

Responsibilities

- Led a multidisciplinary engineering project integrating hardware, embedded systems, sensors, and custom software into a cohesive smart-device solution.
- Architected and developed a Smart-Hub IoT platform leveraging Raspberry Pi and embedded technologies to enable real-time sensor processing, device intelligence, and user interaction capabilities.
- Implemented computer vision-based human detection features using OpenCV, enabling real-time object recognition and automated responses based on sensor input.
- Developed a custom desktop interface using Python and PyQt5 to monitor device activity, visualize system status, and deliver real-time user alerts.
- Programmed embedded components using MicroPython, integrating hardware sensors and enabling communication between edge devices and application software.
- Designed and integrated hardware/software workflows, bridging embedded devices, sensor data acquisition, and user-facing applications to create an end-to-end smart monitoring solution.
- Collaborated across hardware and software disciplines to define system requirements, troubleshoot integration challenges, and deliver a functional prototype.
- Utilized rapid prototyping methodologies, including CAD-driven design and iterative testing, to refine device functionality and improve user experience.

Exas

React Native Mobile Application

Technical Experience

- Languages: Java, TypeScript, JavaScript, SQL
- Frontend: React Native
- Backend: Spring Boot, REST APIs
- Cloud: AWS
- Databases: (Add your database—PostgreSQL, MySQL, DynamoDB, etc.)
- Tools: Git

Responsibilities

- Architected and developed a full-stack Customer Relationship Management (CRM) and Field Service Management (FSM) platform connecting consumers with local businesses through a social media-inspired marketplace for discovering, engaging with, and booking service providers.
- Engineered cross-platform mobile applications for iOS and Android using React Native, delivering a unified user experience for customers and business owners.
- Designed and implemented scalable backend services using Spring Boot, exposing secure REST APIs for user authentication, business management, scheduling, messaging, and customer interactions.
- Built dual-platform functionality supporting both consumer and business workflows, enabling customer discovery, service requests, appointment scheduling, and business marketing through a single application ecosystem.
- Developed CRM capabilities including customer profiles, business directories, service management, appointment history, and communication tools to strengthen customer engagement and retention.
- Implemented Field Service Management (FSM) features including appointment scheduling, calendar management, customer management, notifications, and workflow automation for service-based businesses.
- Leveraged AWS cloud services for application hosting, scalable storage, backend infrastructure, and deployment, designing a cloud-native architecture capable of supporting future growth.
- Designed a modular, scalable system architecture emphasizing maintainability, secure API design, and separation of frontend and backend services.
- Managed the complete software development lifecycle, including product architecture, UI/UX implementation, backend development, database design, cloud deployment, testing, and iterative feature delivery.

REFERENCES

Vishal Srinivasan	Ryan Campbell	Blake Bergstrom	Mike Craven	Mike Hoffman
BNY	4c Strategies	Phoenix Defense	Phoenix Defense	Dignitas Technologies
(689) 808-1378	(954) 309-9158	(609) 602-8991	(407) 539-2975	(407) 601-7847